

MH1200 Linear Algebra I.

The 5th Quiz, 2023.

- Please write down your **name** and **matriculation number** at the top. The duration is **20 minutes**. There are **two** questions. A total of 4 marks.
- You may write on every part of this document and also use extra paper if you need. To get full marks please write down all of your working.

Problem 1:

(2.5 marks.)

- (a) Is it possible to express a vector on the following list as a linear combination of the other vectors?

$$\{(1, 1, -1, 3), (2, 0, 2, 1), (-3, -2, 4, 1), (7, 3, -5, -4)\}.$$

- (b) If it is possible write down such an expression.

- (c) Is this list linearly independent?

$$\begin{bmatrix} 1 & 2 & -3 & 7 \\ 1 & 0 & -2 & 3 \\ -1 & 2 & 4 & -5 \\ 3 & 1 & 1 & -4 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - 3R_1}} \begin{bmatrix} 1 & 2 & -3 & 7 \\ 0 & -2 & 1 & -4 \\ 0 & 4 & 1 & 2 \\ 0 & -5 & 10 & -25 \end{bmatrix} \xrightarrow{\substack{R_3 \rightarrow R_3 + 2R_2 \\ R_4 \rightarrow R_4 - \frac{5}{2}R_2}} \begin{bmatrix} 1 & 2 & -3 & 7 \\ 0 & -2 & 1 & -4 \\ 0 & 0 & 3 & -6 \\ 0 & 0 & \frac{15}{2} & -15 \end{bmatrix}$$

$$\xrightarrow{R_4 \rightarrow R_4 - \frac{15}{6}R_3} \begin{bmatrix} 1 & 2 & -3 & 7 \\ 0 & -2 & 1 & -4 \\ 0 & 0 & 3 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solve for $L_1(1, 1, -1, 3) + L_2(2, 0, 2, 1) + L_3(-3, -2, 4, 1) + L_4(7, 3, -5, -4) = (0, 0, 0, 0)$

Set $L_4 = s, s \in \mathbb{R}$

~~$L_3 = 0$~~

$$3L_3 - 6L_4 = 0$$

$$L_3 = \frac{6s}{3} = 2s$$

$$-2L_2 + L_3 - 4L_4 = 0$$

$$L_2 = \frac{4s + 2s}{-2} = -s$$

$$L_1 = -7s + 6s + 2s = s$$

$$\therefore L_1 = s$$

$$L_2 = -s$$

$$L_3 = 2s$$

$$L_4 = s$$

- c) Since the set of solutions do not only contain the trivial solution $L_1 = L_2 = L_3 = L_4 = 0$, the list is linearly dependent.

a) ~~Yes~~ ~~it is possible~~

b) ~~$(1, 1, -1, 3) - (2, 0, 2, 1) + (7, 3, -5, -4) = -2(-3, -2, 4, 1)$~~

~~$(-3, -2, 4, 1) = -\frac{1}{2}(1, 1, -1, 3) + \frac{1}{2}(2, 0, 2, 1) - \frac{1}{2}(7, 3, -5, -4)$~~

Problem 2:

(2.5 marks.)

Consider vectors

$$\vec{v}_1 = (1, 0, -1, 1), \vec{v}_2 = (2, 1, 0, 2), \vec{v}_3 = (1, 1, -1, 0),$$

and vectors

$$\vec{w}_1 = (0, -a, 0, 5), \vec{w}_2 = (4a, 3, 0, -a).$$

Note that these vectors depend on a parameter a .

For what values of the parameter a is it true that

$$\text{span}\{\vec{w}_1, \vec{w}_2\} \subset \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}.$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4a \\ 0 & 1 & 1 & -a & 3 \\ -1 & 0 & -1 & 0 & 0 \\ 1 & 2 & 0 & 5 & -a \end{array} \right] \xrightarrow{\substack{R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - R_1}} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4a \\ 0 & 1 & 1 & -a & 3 \\ 0 & 2 & 0 & 0 & 4a \\ 0 & 0 & -1 & 5 & -5a \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4a \\ 0 & 1 & 1 & -a & 3 \\ 0 & 0 & -2 & 2a & 4a-6 \\ 0 & 0 & -1 & 5 & -5a \end{array} \right] \xrightarrow{R_4 \rightarrow R_4 - \frac{1}{2}R_3} \left[\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 4a \\ 0 & 1 & 1 & -a & 3 \\ 0 & 0 & -2 & 2a & 4a-6 \\ 0 & 0 & 0 & 5-a & -5a-(2a-3) \\ & & & & = -7a+3 \end{array} \right]$$

For that to be true,

~~$5-a \neq 0$ and $-7a+3 \neq 0$~~

~~$a \neq 5$ and $a \neq \frac{3}{7}$~~

~~$5-a=0$ and $-7a+3=0$~~

~~$a=5$ and $a=\frac{3}{7}$~~

Since there is not one common value of a for both $\vec{w}_1$ and $\vec{w}_2$ for $\vec{w}_1 \in \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ and $\vec{w}_2 \in \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$, there are no values of a for $\text{span}\{\vec{w}_1, \vec{w}_2\} \subset \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$